



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 02

Objective & Lecture Mapping: In Lecture 03, we learned that a mathematically perfect "Table-Driven Agent" is impossible to build due to combinatorial explosion. Instead, we must write Agent Programs. Today, you will implement a **Simple Reflex Agent** using Condition-Action rules, observe its fatal flaws in a partially observable environment, and then upgrade it to a **Model-Based Agent** that maintains an internal memory state.

Part 1: Practical Implementation — Coding the Architectures

Step 1.1: Setting the Trap (Partial Observability)

To see why Simple Reflex agents fail, we must handicap your agent's sensors. We are moving from a Fully Observable world to a Partially Observable one.

1. Open your `visual_grid_game.py` from Lab 01.
2. Locate the `get_percept(self)` method.
3. Modify it so it **no longer returns** the exact global coordinates (`agent_pos`). Instead, it should only return local booleans, e.g., `{ 'wall_ahead': True/False, 'food_here': True/False }` based on checking the adjacent cell in its current facing direction.

Step 1.2: The Simple Reflex Agent (Implementation & Failure)

1. Create a new class called `SimpleReflexAgent` .
2. Implement a `sense_and_act(self, percept)` method that uses strictly IF-THEN logic (Condition-Action rules). *Do not use an `__init__` method to store history.*

Example Logic: IF food_here THEN suck; IF wall_ahead THEN turn_left; ELSE move_forward

3. Run the simulation. **Observe:** Watch the agent get trapped in a corner or a U-shaped wall, infinitely repeating a cycle (e.g., turn left, step forward, hit wall, turn left...).

Step 1.3: The Model-Based Agent (Memory & State)

1. Create a new class called `ModelBasedAgent`.
2. Implement an `__init__(self)` method to initialize an internal memory state (e.g., `self.visited_cells = set()` or a relative movement tracker).
3. In `sense_and_act(self, percept)`, first **update the state** (Transition & Sensor Model) by recording the current percept and the last action taken.
4. Update your IF-THEN rules to query the memory.
Example Logic: IF wall_ahead AND left_is_visited THEN turn_right
5. Run the simulation. **Observe:** The agent should now remember where it has been, realize it is in a loop, and choose an alternate path to escape.

Part 2: Theoretical Evaluation & Lecture Mapping

Based on your practical observations above, answer the following questions to map your code directly to the architectural theories covered in Lecture 02.

1. (Remember) According to Lecture 02, why is it impossible to program a mathematically perfect "Table-Driven Agent" for complex environments like Chess? What happens as the agent's lifetime increases?

A table-driven agent requires storing a pre-computed action for every possible sequence of percepts, making it mathematically impossible for complex environments like Chess because the lookup table size grows exponentially with the environment's complexity and time horizon. As the agent's lifetime increases, the state space explodes beyond the physical storage capacity of any computing system, causing memory failure and making lookups computationally intractable.

2. (Understand) Look at the code you wrote for your `SimpleReflexAgent` . Identify and explain the specific lines of code that represent the "Condition-Action Rules" discussed in the lecture.

In the `SimpleReflexAgent` implementation, the condition-action rules are represented by `if percept.get('wall_ahead', False): return random.choice(['Left', 'Right', 'Down'])` and `return 'Up'`. Here, `percept.get('wall_ahead', False)` serves as the condition evaluating the current immediate sensor input, while returning a specific movement directional string represents the mapped action dictated by the agent's hardcoded lookup behavior.

3. (Analyze) Your `SimpleReflexAgent` likely got stuck in an infinite loop during Step 1.2. Based on the lecture, analyze exactly why this happened. How did the combination of "Partial Observability" and a lack of "Percept History" cause this failure?

The `SimpleReflexAgent` got trapped in an infinite loop because its decision-making relies strictly on the current percept without any memory of previous actions or visited cells. When placed in a U-shaped wall under partial observability, the agent receives identical local sensor inputs (`wall_ahead: True`) at the same position step after step; because it lacks a percept history to recognize that it has repeatedly visited that exact location, it deterministically or randomly executes the same reflexive turning rules continuously.

4. (Evaluate) In Step 1.3, you added an internal state to your `ModelBasedAgent` . Evaluate how your specific code handles the "Transition Model" (how the world evolves) and the "Sensor Model" (how the agent's actions affect the world).

In `ModelBasedAgent`, the Transition Model is handled where the internal state updates relative coordinates `self.current_pos = (x, y)` based on `self.last_action` assuming the world remains static, while the Sensor Model is reflected when `percept.get('hit_wall', False)` and `percept.get('wall_ahead', False)` correct this expectation by determining whether the attempted action actually succeeded or was blocked by the environment.

Finalize and Lock Lab 02 Documentation

Incomplete: 0/11 questions answered. Please provide detailed responses for all questions.



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